

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	FSD Workshop-I	25VA351	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1. To introduce students to server-side development concepts including the architecture, core modules, and event-driven programming model of Node.js.		
2. To familiarize students with asynchronous programming techniques and file system operations for efficient handling of files and directories in Node.js		
3. To enable students to develop web servers and implement routing mechanisms using HTTP modules and Express.js.		
4. To provide knowledge of NoSQL database concepts and data modeling using MongoDB for performing CRUD operations.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	1. Create a custom EventEmitter that triggers "greet" or "exit" 2. Simulate DOM-like event handling in Node.js using events 3. Visualize the event loop using setTimeout, setImmediate, and process.nextTick	2
2	Perform CRUD operations on files using fs module	2
3	Create a basic HTTP server using http.createServer(), Respond with “Hello World” and return headers + status code	2
4	Set up a basic GraphQL API using or express-graphql	2
5	Create routes that handle GET, POST, PUT, DELETE in Node.js	2
6	Create a simple Express.js server that: Serves an HTML file using the fs module.Has routes like /, /about, and /contact. Each route should serve a separate HTML file.	2
7	Install MongoDB locally (or use MongoDB Atlas). Create a new database named studentDB with a collection students	2
8	Perform the following on studentDB.students collection: Insert 5 student records. Find all students older than 20. Update a student’s age using \$set. Delete one student.	2

9	Create a REST API using Express and MongoDB: POST /students – Add a new student GET /students – Retrieve all students PUT /students/:id – Update a student by ID DELETE /students/:id – Delete a student	4
10	Push your MongoDB + Express project to GitHub. Ensure: Proper README.md .gitignore included Folder structure is clean	4
Exp. No.	Value Added Experiments	LI Hours
11	Dockerize your Node.js + MongoDB app: Write a Dockerfile, Build and run the Docker image locally	2
12	Create a Student Management System(SMS) applications using react, express, and mongo DB then Dockerize and deploy on render	4

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Web Technologies – A Developer’s Perspective	N.P. Gopalan and J. Akilandeswari	3rd	Pearson	2025
2	Node.js, Express, MongoDB & JavaScript	William Stallings	11th	Pearson	2024
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Learning Web Design	Jennifer Robbins	5th	Mc Graw Hill	2012
2	Web Development with Node and Express	Ethan Brown	8th	"O'Reilly Media	2014
Web References					
1. AWS Academy – https://aws.amazon.com/training/ 2. Microsoft Learn – https://learn.microsoft.com 3. Google Cloud Skills Boost – https://cloudskillsboost.google 4. Coursera – https://www.coursera.org					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand the architecture, core modules, and event-driven programming model of Node.js for building scalable server-side applications.	K2
CO2	Apply asynchronous programming concepts and Node.js file system modules to perform file and directory operations efficiently.	K3
CO3	Develop web servers and implement routing mechanisms using HTTP modules and Express.js to handle client requests and responses.	K3
CO4	Develop web servers and implement routing mechanisms using HTTP modules and Express.js to handle client requests and responses.	K3
CO5	Design, develop, and deploy RESTful APIs integrating Node.js, Express.js, and MongoDB, and test the APIs using Postman with version control using GitHub.	K6

KL- Bloom’s Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	1	-	2						1			
CO2	3	3	2	1	3						1			
CO3	3	3	3	2	3				1	1	1			
CO4	3	3	3	2	3			1	2		2			
CO5	3	2	3	2	3			1	2	2	3			
Course Avg.	3.00	2.60	2.40	1.40	2.80			0.40	1.00	0.60	1.60			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50